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Data Structures and Algorithms
Merge sort

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Objectives

- Divide and conquer
- Merge sort
- Merge sort- algorithm and example

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DIVIDE AND CONQUER

- Divide and Conquer-an algorithm design technique.
  - Divide the Original Problem into subproblems.
  - Solve every subproblems individually and then combine these solutions to create a solution to the original problem.

DIVIDE AND CONQUER

- Divide the problem into several subproblems that are smaller instances of the same problem.
- Conquer these subproblems by solving them recursively.
- Combine the solutions to the subproblems into the solution for the original problem.

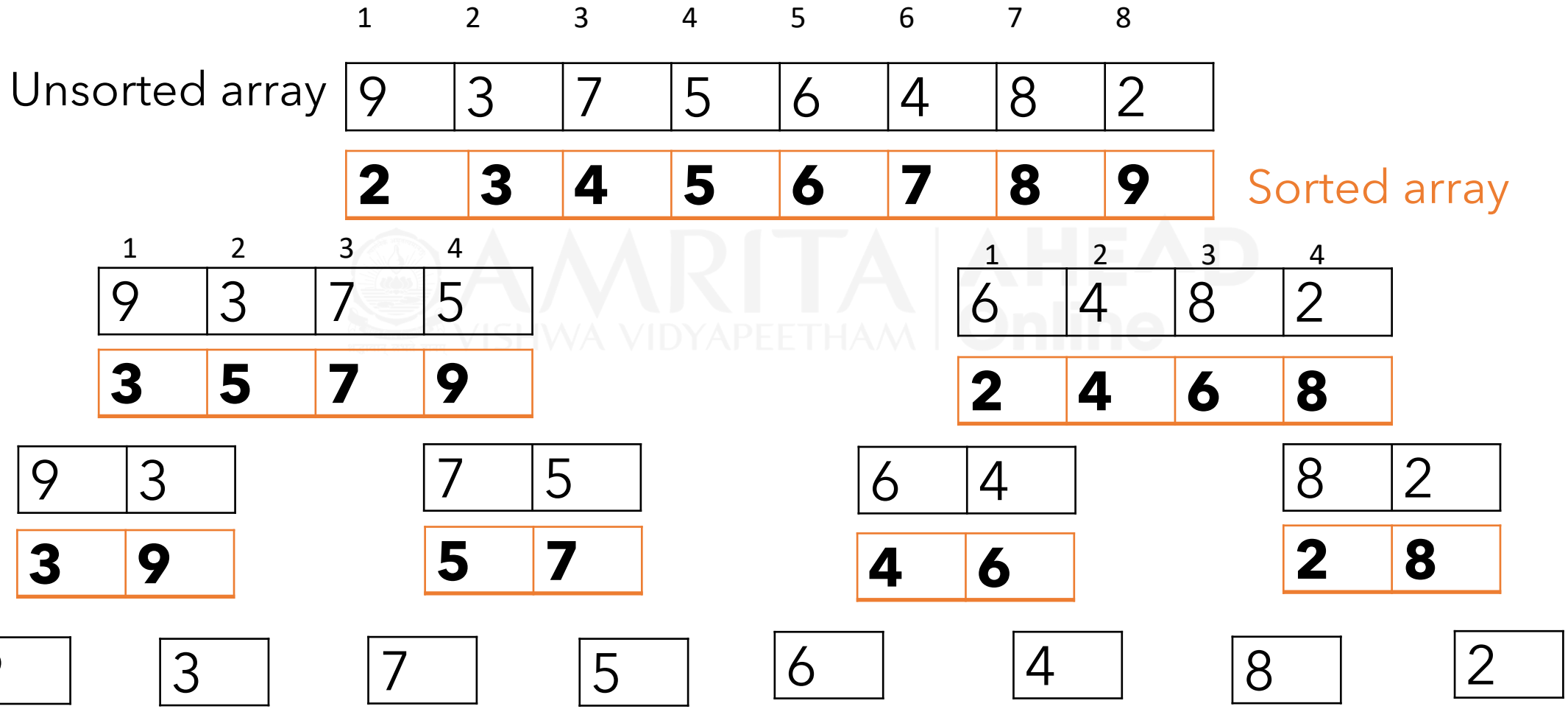
MERGE SORT

- The merge sort algorithm follows the divide-and-conquer strategy.

It operates as follows:

- Divide: Divide the n-element sequence to be sorted into two subsequences.
- Conquer: Recursively sort the two subsequences using merge sort algorithm.
- Combine: Merge the two sorted subsequences to get a sorted sequence.

Merge Sort Example



MERGE SORT-ALGORITHM

MERGE-SORT

```
1 MERGE-SORT(A, l,n):
2     if l < n then
3         m = ⌊(l + n)/2⌋
4         MERGE-SORT(A,l,m)
5         MERGE-SORT(A, m + 1,n)
6         MERGE(A, l,m,n)
7     end
8 end
```

The procedure MERGE-SORT(A,l,n) sorts the elements in the array A[l..n]

l and n are the start and end indices.

If l >=n the subarray has at most one element and is therefore already sorted.

Otherwise, the divide step computes an index m that partitions A[l..n] into two sub arrays: A[l..m] and A[m+1..n]

The MERGE procedure can be used as a subroutine in the merge sort algorithm to merge the sorted sub arrays to a single sorted array.

MERGE ALGORITHM

MERGE

- The main operation in merge sort algorithm is the merging of two sorted arrays or sequences.

- Merging is done by calling an auxiliary procedure *MERGE (A, l, m, n)* *where,*
  A is an array and l, m, n are the start, mid and end indices or positions of array respectively (l<= m < n).

- *The MERGE (A, l, m, n) procedure merge the subarrays A[l..m] and A[m+1..n] by assuming that the sub arrays are in sorted order.*

- Form a single sorted array that replaces the existing array *A[l..n].*

MERGE ALGORITHM

MERGE

```
1  MERGE(A, l, m, n):
2      p1 = m − l+1
3      p2 = n − m
4      Let L[1..p1+1] and R[1..p2+1] are new arrays
5      for i = 1 to p1 do
6          L[i]=A[l+i-1]
7      end
8      for j = 1 to p2 do
9          R[j]=A[m+j]
10     end
11     L[p1+1]=∞
12     R[p2+1]=∞
13     i=1
14     j=1
15     for k =l to n do
16         if L[i] <=R[j] then
17             A[k] =L[i]
18             i=i + 1
19         else
20             A[k] =R[j]
21             j=j + 1
22     end
23
   end
```

- P1 -length of the subarray A[l..m]
- p2 - length of the subarray A[m+1..n]
- We create arrays L and R (“left” and “right”), of lengths p1+1 and p2+1, respectively
- The extra position in each array will hold the sentinel.
- The for loops copies the subarray A[l:m] to L[1..p1] and the sub array A[m+1..n] to R[1..p2]
- Put the sentinels at the ends of the arrays L and R.
- i= index of left subarray
- j= index of right subarray
- k=index of current array where resultant sorted values stored.
- Initializing I and j as 1 and start loop iteration.

MERGE ALGORITHM

Example



A=

2	3	4	5	6	7	8	9

```
1  MERGE(A, l, m, n):
2      p1 = m − l+1
3      p2 = n − m
4      Let L[1..p1+1] and R[1..p2+1] are new arrays
5      for i = 1 to p1 do
6          L[i]=A[l+i-1]
7      end
8      for j = 1 to p2 do
9          R[j]=A[m+j]
10     end
11     L[p1+1]=∞
12     R[p2+1]=∞
13     i=1
14     j=1
15     for k =l to n do
16         if L[i] <=R[j] then
17             A[k] =L[i]
18             i=i + 1
19         else
20             A[k] =R[j]
21             j=j + 1
22     end
23
   end
```

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Summary

Introduced Merge Sort as example for Divide and Conquer paradigm.

Discussed the process of merging and sorting with examples and algorithms.

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References

Introduction to Algorithms- Thomas H Cormen.

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